

SF1811 Optimization

Lecture 12: Nonlinear optimization with linear constraints

A note on optimality conditions, Part A

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Nonlinear optimization with linear constraints

Main question:

How does the absence of a feasible direction with strict first-order decrease lead to Lagrange multipliers?

Today:

- Active constraints and feasible directions.
- The problem of finding a feasible direction with strict first-order decrease.
- An associated primal–dual pair of linear programs.
- First-order necessary optimality conditions for linear constraints.
- Complementarity and the role of the multipliers.
- Sufficiency for convex problems.

The linearly constrained problem

Consider

$$(P_{\text{lin}}) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & b_i - a_i^{\mathsf{T}} x = 0, \quad i \in \mathcal{E}, \\ & b_i - a_i^{\mathsf{T}} x \leq 0, \quad i \in \mathcal{I}, \end{array}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and

$$\mathcal{E} \cap \mathcal{I} = \emptyset, \qquad \mathcal{E} \cup \mathcal{I} = \{1, \ldots, m\}.$$

This problem includes:

- equality-constrained problems when $\mathcal{I} = \emptyset$;
- inequality-constrained problems when $\mathcal{E} = \emptyset$;
- unconstrained problems when $\mathcal{E} = \mathcal{I} = \emptyset$.

The objective may be nonlinear, but the constraints are linear.

Active inequality constraints

Let $\hat{x}$ be feasible to (P_{lin}).

An inequality constraint is active at $\hat{x}$ if

$$b_i - a_i^T \hat{x} = 0.$$

The active index set is

$$\mathcal{I}_a(\hat{x}) = \left\{ i \in \mathcal{I} : b_i - a_i^T \hat{x} = 0 \right\}.$$

For an inactive inequality constraint,

$$b_i - a_i^T \hat{x} < 0.$$

Thus there is some room before that constraint can be violated.

Only active inequalities restrict sufficiently small feasible steps to first order.

Descent directions

A vector $d \in \mathbb{R}^n$ is a descent direction for f at $\hat{x}$ if

$$f(\hat{x} + td) < f(\hat{x})$$

for every positive t sufficiently close to zero.

If

$$\nabla f(\hat{x})d < 0,$$

then d is a descent direction.

Indeed,

$$f(\hat{x} + td) = f(\hat{x}) + t\nabla f(\hat{x})d + o(t),$$

and the negative linear term dominates for positive t sufficiently close to zero.

The gradient identifies directions of local decrease.

Feasible directions for linear constraints

Consider a step

$$x(t) = \hat{x} + td, \qquad t > 0.$$

For an equality constraint,

$$b_i - a_i^\mathsf{T} x(t) = b_i - a_i^\mathsf{T} \hat{x} - t a_i^\mathsf{T} d.$$

Since $b_i - a_i^\mathsf{T} \hat{x} = 0$, equality is preserved if and only if

$$a_i^\mathsf{T} d = 0, \qquad i \in \mathcal{E}.$$

For an active inequality constraint,

$$b_i - a_i^\mathsf{T} x(t) = -t a_i^\mathsf{T} d.$$

It remains satisfied for $t > 0$ if and only if

$$a_i^\mathsf{T} d \geq 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

What about inactive constraints?

For $i \in \mathcal{I} \setminus \mathcal{I}_a(\hat{x})$,

$$b_i - a_i^\top \hat{x} < 0.$$

Along $x(t) = \hat{x} + td$,

$$b_i - a_i^\top x(t) = b_i - a_i^\top \hat{x} - t a_i^\top d.$$

If $a_i^\top d \geq 0$, the constraint remains satisfied for every $t \geq 0$. If $a_i^\top d < 0$, it remains satisfied whenever

$$0 \leq t < \frac{-(b_i - a_i^\top \hat{x})}{-a_i^\top d}.$$

The upper bound is strictly positive. Hence inactive constraints impose no additional condition on a sufficiently small step.

For linear constraints, the feasible-direction conditions are exact.

No feasible direction with strict first-order decrease

If $\hat{x}$ is a local minimizer, there cannot exist a vector d satisfying

$$\nabla f(\hat{x})d < 0,$$

$$a_i^T d = 0, \qquad i \in \mathcal{E},$$

and

$$a_i^T d \geq 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

Otherwise, d would be both

- a descent direction for the objective;
- a feasible direction for the constraints.

Then

$$f(\hat{x} + td) < f(\hat{x})$$

for feasible points $\hat{x} + td$ arbitrarily close to $\hat{x}$, contradicting local optimality.

The feasible-direction problem

The search for a feasible direction with strict first-order decrease can be formulated as the linear program

$$
(P_{\text{dir}}) \qquad \begin{array}{lll} \text{minimize}_d & \nabla f(\hat{x})d & \\ \text{subject to} & a_i^{\mathsf{T}} d = 0, & i \in \mathcal{E}, \\ & a_i^{\mathsf{T}} d \geq 0, & i \in \mathcal{I}_a(\hat{x}). \end{array}
$$

The point

$$
d = 0
$$

is feasible, so the optimal value is at most zero.

There are two possibilities:

- the optimal value is 0, and no feasible direction with strict first-order decrease exists;
- the problem is unbounded below, and a feasible direction with strict first-order decrease exists.

Why are there only two possibilities?

Suppose that a feasible d satisfies

$$\nabla f(\hat{x})d < 0.$$

For every $t > 0$, the vector td is also feasible to (P_{dir}), because

$$a_i^{\mathsf{T}}(td) = 0, \qquad i \in \mathcal{E},$$

and

$$a_i^{\mathsf{T}}(td) \geq 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

Moreover,

$$\nabla f(\hat{x})(td) = t\nabla f(\hat{x})d \longrightarrow -\infty$$

as $t \to \infty$.

Thus a single feasible direction d satisfying $\nabla f(\hat{x})d < 0$ makes (P_{dir}) unbounded below.

At a local minimizer, the optimal value of (P_{dir}) must be zero.

The dual direction problem

The dual of (P_{dir}) is

$$(D_{\text{dir}}) \qquad \begin{aligned} &\text{maximize} && 0 \\ &\text{subject to} && \sum_{i\in\mathcal{E}} a_i u_i + \sum_{i\in\mathcal{I}_a(\hat{x})} a_i y_i = \nabla f(\hat{x})^{\mathsf{T}}, \\ & && y_i \geq 0, \quad i \in \mathcal{I}_a(\hat{x}), \end{aligned}$$

where

$$u_i \text{ is free}, \qquad i \in \mathcal{E}.$$

The dual objective is zero because all right-hand sides in (P_{dir}) are zero.

Strong LP duality gives multipliers

Assume that $\hat{x}$ is a local minimizer. Then (P_{dir}) has optimal value zero.

By strong duality for linear programming, (D_{dir}) has a feasible solution. Hence there exist multipliers $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i \geq 0$, $i \in \mathcal{I}_a(\hat{x})$, such that

$$\nabla f(\hat{x})^{\mathsf{T}} = \sum_{i \in \mathcal{E}} a_i \hat{u}_i + \sum_{i \in \mathcal{I}_a(\hat{x})} a_i \hat{y}_i.$$

Equivalently,

$$\nabla f(\hat{x}) - \sum_{i \in \mathcal{E}} \hat{u}_i a_i^{\mathsf{T}} - \sum_{i \in \mathcal{I}_a(\hat{x})} \hat{y}_i a_i^{\mathsf{T}} = 0.$$

The multipliers are dual variables for the feasible-direction problem.

Adding the inactive constraints

For every inactive inequality constraint, define

$$\hat{y}_i = 0, \qquad i \in \mathcal{I} \setminus \mathcal{I}_a(\hat{x}).$$

Then stationarity can be written using all inequality constraints:

$$\nabla f(\hat{x}) - \sum_{i \in \mathcal{E}} \hat{u}_i a_i^\mathsf{T} - \sum_{i \in \mathcal{I}} \hat{y}_i a_i^\mathsf{T} = 0.$$

Furthermore,

$$\hat{y}_i (b_i - a_i^\mathsf{T} \hat{x}) = 0, \qquad i \in \mathcal{I}.$$

Indeed:

- if constraint i is active, then $b_i - a_i^\mathsf{T} \hat{x} = 0$;
- if constraint i is inactive, then $\hat{y}_i = 0$.

This is the complementarity condition.

First-order necessary optimality conditions for linear constraints

$\hat{x}$ is a local minimizer

$\Downarrow$

No feasible direction with strict first-order decrease exists

$\Updownarrow$

$\text{optval}(P_{\text{dir}}) = 0$

$\Updownarrow$ by strong LP duality

$\text{optval}(D_{\text{dir}}) = 0$

$\Updownarrow$

Lagrange multipliers exist

First-order necessary optimality conditions for linear constraints

Theorem (KKT conditions for linear constraints)

If $\hat{x}$ is a local minimizer of (P_{lin}), then there exist multipliers $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{I}$, such that:

1. $\nabla f(\hat{x}) - \sum_{i \in \mathcal{E}} \hat{u}_i a_i^{\mathsf{T}} - \sum_{i \in \mathcal{I}} \hat{y}_i a_i^{\mathsf{T}} = 0.$ *(Stationarity)*
2. $b_i - a_i^{\mathsf{T}} \hat{x} = 0, \quad i \in \mathcal{E}, \qquad b_i - a_i^{\mathsf{T}} \hat{x} \leq 0, \quad i \in \mathcal{I}.$ *(Primal feasibility)*
3. $\hat{y}_i \geq 0, \quad i \in \mathcal{I},$ *(Dual feasibility)*
4. $\hat{y}_i (b_i - a_i^{\mathsf{T}} \hat{x}) = 0, \quad i \in \mathcal{I}.$ *(Complementarity)*

The Lagrangian

Define

$$L(x, u, y) = f(x) + \sum_{i \in \mathcal{E}} u_i (b_i - a_i^\mathsf{T} x) + \sum_{i \in \mathcal{I}} y_i (b_i - a_i^\mathsf{T} x).$$

Then

$$\nabla_x L(x, u, y) = \nabla f(x) - \sum_{i \in \mathcal{E}} u_i a_i^\mathsf{T} - \sum_{i \in \mathcal{I}} y_i a_i^\mathsf{T}.$$

Thus stationarity is

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0.$$

The KKT conditions for linear constraints consist of:

- stationarity of the Lagrangian;
- primal feasibility;
- nonnegativity of the inequality multipliers;
- complementarity.

Geometric interpretation of stationarity

At a point satisfying stationarity,

$$\nabla f(\hat{x})^{\mathsf{T}} = \sum_{i \in \mathcal{E}} a_i \hat{u}_i + \sum_{i \in \mathcal{I}_a(\hat{x})} a_i \hat{y}_i.$$

Thus the objective gradient is balanced by normals to the active constraints.

- Equality multipliers are free because either side of an equality constraint is forbidden.
- Inequality multipliers are nonnegative because active inequalities restrict motion only toward one side.
- Inactive inequalities have zero multipliers because they do not restrict sufficiently small steps.

A one-dimensional example

Consider

$$\begin{aligned} &\text{minimize} && \tfrac{1}{2}x^2 \\ &\text{subject to} && 1 - x \leq 0. \end{aligned}$$

The first-order conditions are

$$\hat{x} - \hat{y} = 0,$$

$$1 - \hat{x} \leq 0, \qquad \hat{y} \geq 0, \qquad \hat{y}(1 - \hat{x}) = 0.$$

If the constraint is inactive, then $\hat{y} = 0$, and stationarity gives $\hat{x} = 0$, which is infeasible.

Hence the constraint is active:

$$\hat{x} = 1.$$

Stationarity then gives

$$\hat{y} = 1.$$

Thus the solution is

$$\boxed{\hat{x} = 1, \ \hat{y} = 1.}$$

Equality and inequality constraints differ

Compare

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^2 \\ \text{subject to} & 1 - x = 0 \end{array} \quad \text{and} \quad \begin{array}{ll} \text{minimize} & \frac{1}{2}x^2 \\ \text{subject to} & 1 - x \leq 0. \end{array}$$

For the equality-constrained problem:

$$\hat{x} - \hat{u} = 0, \qquad 1 - \hat{x} = 0,$$

so

$$\hat{x} = 1, \qquad \hat{u} = 1.$$

For the inequality-constrained problem, the same point is optimal because the corresponding multiplier is nonnegative:

$$\hat{y} = 1 \geq 0.$$

If an equality multiplier has the wrong sign for the corresponding inequality, the equality-constrained solution need not solve the inequality- constrained problem.

Convexity makes the KKT conditions sufficient

Assume that f is convex and that $\hat{x}$, $\hat{u}$, and $\hat{y}$ satisfy the KKT conditions for linear constraints.

For any feasible x, convexity gives

$$f(x) \geq f(\hat{x}) + \nabla f(\hat{x})(x - \hat{x}).$$

Using stationarity,

$$\nabla f(\hat{x}) = \sum_{i \in \mathcal{E}} \hat{u}_i a_i^{\mathsf{T}} + \sum_{i \in \mathcal{I}} \hat{y}_i a_i^{\mathsf{T}}.$$

The equality terms vanish between feasible points. For each inequality, feasibility, dual feasibility, and complementarity give

$$\hat{y}_i a_i^{\mathsf{T}}(x - \hat{x}) \geq 0.$$

Hence

$$f(x) \geq f(\hat{x}).$$

For a convex objective and linear constraints, the KKT conditions certify global optimality.

Looking ahead: nonlinear constraints

For linear constraints,

$$b_i - a_i^T(\hat{x} + td) = b_i - a_i^T\hat{x} - ta_i^T d.$$

There is no approximation or remainder term.

For a nonlinear constraint $g_i(x) \leq 0$,

$$g_i(\hat{x} + td) = g_i(\hat{x}) + t\nabla g_i(\hat{x})d + o(t).$$

The linearized condition

$$\nabla g_i(\hat{x})d \leq 0$$

need not by itself guarantee the existence of nearby feasible points in that direction.

Next lecture:

When does the linearized feasible-direction problem describe the true local geometry well enough?

Main ideas

After this lecture, you should be able to:

- identify active inequality constraints;
- characterize feasible directions for linear constraints;
- explain why a local minimizer admits no feasible direction with strict first-order decrease;
- formulate the feasible-direction problem as a linear program;
- derive the KKT conditions for linear constraints using strong LP duality;
- state stationarity, primal feasibility, dual feasibility, and complementarity;
- interpret the roles of equality and inequality multipliers;
- explain why the KKT conditions for linear constraints are sufficient for global optimality when the objective is convex.

For linear constraints, Lagrange multipliers are dual variables for the problem of finding a feasible direction with strict first-order decrease.